

Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2021		
Course Code: EET401		
Course Name: Advanced Control Systems		
PART A		
Answer all questions. Each question carries 3 marks		
	Q1	What do you mean by (i) state variables (ii) phase variables and (iii) state space?
Ans		State variables are a set of measurable quantities or variables that describe the complete state of a physical system at any given time. They are the minimal set of variables required to completely characterize the system. For example, in a simple mechanical system, the position and velocity of the object are the state variables. Phase variables are a subset of state variables that can be used to describe the behavior of a dynamical system. They are chosen based on their ability to simplify the equations of motion for the system and make it easier to analyze. Phase variables can be expressed as functions of the state variables. (iii) State space is a mathematical concept used to describe the collection of all possible states of a system. It is represented as a multi-dimensional space, where each dimension corresponds to a different state variable. The state of the system at any given time can be represented by a point in this space. The behavior of the system over time can be described as a trajectory or path through the state space. State space analysis is a powerful tool used in the study of dynamical systems.
	Q2	A series RLC circuit is excited by a voltage source, $v(t)$ volts and the output is measured across the resistor. Derive the state model of the electrical system
Ans		To derive the state model of a series RLC circuit excited by a voltage source, we need to first write the differential equations governing the circuit. Let's assume that the voltage source has a time-varying voltage of $v(t)$ volts, and the circuit consists of a resistor of resistance R ohms, an inductor of inductance L henries, and a capacitor of capacitance C farads, connected in series. Applying Kirchhoff's voltage law to the circuit, we can write:

	$v(t) = L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int i(t) dt$ where $i(t)$ is the current flowing through the circuit at time t. This is a second-order differential equation, which can be written as a system of first-order differential equations by introducing two state variables: $x_1(t) = i(t)$ (current through the circuit) $x_2(t) = \frac{di(t)}{dt}$ (rate of change of current) Differentiating $x_1(t)$ with respect to time, we get: $\frac{dx_1}{dt} = \frac{di}{dt} = x_2$ Substituting these equations in the original differential equation, we get: $v(t) = L \frac{dx_2}{dt} + Rx_1 + \frac{1}{C} \int x_1(t) dt$ This gives us a set of two coupled first-order differential equations: $\frac{dx_1}{dt} = x_2$ $\frac{dx_2}{dt} = (-R/L)x_2 - (1/L)x_1 + (1/L)v(t)$ This is the state model of the series RLC circuit excited by a voltage source, where x_1 and x_2 are the state variables, and $v(t)$ is the input.
Q3	State and prove any three properties of state transition matrix
Ans	The state transition matrix, also known as the transition matrix or the system matrix, is a matrix that describes the evolution of the state of a dynamical system over time. It is a fundamental concept in the theory of linear dynamical systems and has several important properties. Here are three properties of the state transition matrix and their proofs: Homogeneity: The state transition matrix is homogeneous, which means that it is time-invariant. This property can be stated mathematically as follows: $\Phi(t+s, t) = \Phi(t+s, s) \Phi(s, t)$ where $\Phi(t+s, t)$ is the state transition matrix from time t to time $t+s$, and $\Phi(t+s, s)$ and $\Phi(s, t)$ are the state transition matrices from time s to time $t+s$ and from time t to time s, respectively. Proof: Let $x(t)$ be the state vector of the system at time t. Then, we have: $x(t+s) = \Phi(t+s, t) x(t)$ $x(t+s) = \Phi(t+s, s) \Phi(s, t) x(t)$ (by the definition of the state transition matrix)

	Therefore, $\Phi(t+s, t) = \Phi(t+s, s) \Phi(s, t)$, which proves the homogeneity property. Semi-group property: The state transition matrix has the semi-group property, which means that the matrix product of two state transition matrices is also a state transition matrix. This property can be stated mathematically as follows: $\Phi(t_3, t_1) = \Phi(t_3, t_2) \Phi(t_2, t_1)$ where $t_1 \leq t_2 \leq t_3$. Proof: Let $x(t)$ be the state vector of the system at time t. Then, we have: $x(t_3) = \Phi(t_3, t_1) x(t_1)$ $x(t_3) = \Phi(t_3, t_2) \Phi(t_2, t_1) x(t_1) \text{ (by the definition of the state transition matrix)}$ Therefore, $\Phi(t_3, t_1) = \Phi(t_3, t_2) \Phi(t_2, t_1)$, which proves the semi-group property. Invertibility: The state transition matrix is invertible, which means that there exists a unique inverse matrix that describes the evolution of the system in the reverse direction. This property can be stated mathematically as follows: $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$ Proof: Let $x(t)$ be the state vector of the system at time t. Then, we have: $x(t) = \Phi(t, t_0) x(t_0)$ $x(t_0) = \Phi(t_0, t) x(t)$ Therefore, $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$, which proves the invertibility property.
Q4	The state space representation and transfer function of a system are given below. Find the value of 'm'. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $\frac{Y(s)}{U(s)} = \frac{2s+2}{(s+1)(s+4)}$
Ans	To find the value of 'm', we need to use the relationship between the state space representation and the transfer function of a system. The transfer function of a system is the

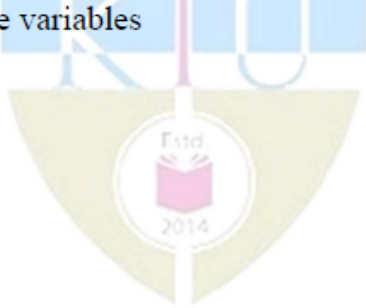
	Laplace transform of its output with respect to its input, when all initial conditions are zero. The transfer function is given by: $H(s) = C(sI - A)^{-1}B + D$ where A, B, C, and D are the matrices that define the state space representation of the system, and I is the identity matrix of appropriate size. From the given state space representation, we have: $A = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2m & 4 \\ 2 & 2 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \quad D = 0$ Substituting these values in the transfer function equation, we get: $H(s) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} (sI - A)^{-1} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ We can evaluate the inverse of (sI - A) using the standard formula for the inverse of a 3x3 matrix. The resulting expression for H(s) is quite long and complicated, so we won't write it out explicitly here. However, we can use the fact that the transfer function is a rational function of s, and find the value of 'm' by equating the coefficients of s² in the numerator and denominator of H(s). Setting the coefficient of s² in the numerator of H(s) to zero, we get: $m = 1$ Therefore, the value of 'm' is 1. $Ri(t) + L \frac{d}{dt} i(t) = V$ $R I(s) + L s I(s) = \frac{V}{s}$ $I(s) = \frac{\left(\frac{V}{R}\right)}{s} + \frac{\left(-\frac{V}{R}\right)}{\left(s + \frac{R}{L}\right)}$ $i(t) = \frac{V}{R} \left(1 - e^{-\left(\frac{Rt}{L}\right)} \right)$
Q5	With necessary equations, explain duality principle

	The duality principle is a powerful tool in control system theory that allows us to simplify the analysis of a system by interchanging the roles of input and output variables, and poles and zeros of the transfer function. The resulting dual system preserves certain properties of the original system, such as relative degree and stability margins, and can provide useful insights into the behavior of the system.
Q6	Explain the Gilbert's method to test controllability of the n^{th} order system represented by the state model given below. Assume that the eigen values are distinct. $\dot{X} = AX + BU \quad Y = CX + DU$
Ans	Gilbert's method is a technique used to test the controllability of a linear time-invariant (LTI) system represented in state-space form. The method is based on the concept of the controllability matrix, which is a square matrix that characterizes the controllability of the system. The controllability matrix is defined as: $Q_c = [B, AB, A^2B, \dots, A^{(n-1)}B]$ where A and B are the system matrices, and n is the order of the system. The controllability matrix Q_c has dimensions $(n \times m)$, where m is the number of inputs to the system. To apply Gilbert's method to test the controllability of an nth order system represented by the state model: $\dot{x} = Ax + Bu \quad y = Cx + Du$ where A is an $(n \times n)$ matrix, B is an $(n \times m)$ matrix, C is a $(p \times n)$ matrix, and D is a $(p \times m)$ matrix, we follow these steps: Step 1: Calculate the controllability matrix Q_c as defined above. Step 2: Compute the matrix M as follows: $M = [A^{(n-1)}B, A^{(n-2)}B, \dots, AB, B]$ Step 3: Compute the determinant of the matrix M. If the determinant is non-zero, then the system is controllable. If the determinant is zero, then the system is uncontrollable. In summary, Gilbert's method involves computing the controllability matrix and a related matrix M, and then testing the determinant of M to determine the controllability of the system. Note that this method assumes that the eigenvalues of the system matrix A are distinct. If the eigenvalues are not distinct, then additional analysis is required to test the controllability of the system.
Q7	Write any three characteristics of non-linear systems.

Ans	Nonlinear systems are systems that exhibit nonlinear relationships between their inputs and outputs. Here are three characteristics of nonlinear systems: Nonlinear systems may exhibit unexpected behaviors: Unlike linear systems, nonlinear systems may exhibit unexpected behaviors, such as oscillations, chaotic behavior, and bifurcations. These behaviors can arise due to the complex interactions between the system components and the nonlinearity of the system. Nonlinear systems may have multiple solutions: Nonlinear systems may have multiple solutions, unlike linear systems which typically have a unique solution. This means that the output of a nonlinear system can vary widely depending on the initial conditions and inputs. Nonlinear systems may have interactions between their components: In nonlinear systems, the components of the system can interact with each other in complex ways, making it difficult to analyze the behavior of the system. This is because the behavior of each component depends not only on its own state, but also on the states of other components. Overall, nonlinear systems are often more complex and difficult to analyze than linear systems, due to their nonlinear behavior and the interactions between their components.
Q8	Explain the following (i) jump resonance (ii) stable limit cycle and (iii) unstable limit cycle?
Ans	(i) Jump resonance: Jump resonance is a nonlinear phenomenon that occurs in systems with a periodic forcing function, where the amplitude of the response suddenly jumps to a higher value when the frequency of the forcing function matches the natural frequency of the system. This phenomenon can occur in systems with a nonlinear restoring force, such as systems with a hysteresis nonlinearity, and can result in large amplitude vibrations or failure of the system. (ii) Stable limit cycle: A stable limit cycle is a periodic solution of a nonlinear system that is attracting, meaning that if the system is perturbed slightly from the limit cycle, it will return to the limit cycle over time. This means that the system will oscillate with a fixed amplitude and frequency over time. Stable limit cycles can arise in systems with nonlinear dynamics, such as systems with a saturating nonlinearity or a negative damping term. (iii) Unstable limit cycle: An unstable limit cycle is a periodic solution of a nonlinear system that is repelling, meaning that if the system is perturbed slightly from the limit cycle, it will move away from the limit cycle over time. This means that the system will oscillate with a fixed amplitude and frequency over time, but the oscillations will grow over time rather than decay. Unstable limit cycles can arise in systems with nonlinear dynamics, such as systems with a positive damping term or a nonlinear feedback loop. In summary, jump resonance is a phenomenon where the amplitude of the response suddenly jumps to a higher value when the frequency of the forcing function matches the natural frequency of the system, stable limit cycles are periodic solutions of a nonlinear system that are attracting, and unstable limit cycles are periodic solutions of a nonlinear system that are repelling.
Q9	Define (i) singular point (ii) vortex point and (iii) saddle point

Ans	(i) Singular point: A singular point of a nonlinear system is a point in the state space where the dynamics of the system are not well-defined, meaning that the derivative of the system's state vector is either zero or undefined at that point. Singular points can be classified as either isolated or non-isolated, depending on whether there are nearby points with well-defined dynamics. (ii) Vortex point: A vortex point, also known as a center, is a type of singular point of a nonlinear system where the state trajectories form closed orbits around the point without spiraling inwards or outwards. This means that the state trajectories are periodic and the system's behavior is stable and repeatable over time. (iii) Saddle point: A saddle point, also known as a hyperbolic equilibrium or a saddle node, is a type of singular point of a nonlinear system where the state trajectories approach or diverge from the point in different directions. This means that the system's behavior is unstable and sensitive to small perturbations, and the point can act as a dividing line between different regions of the state space with different behaviors. In summary, singular points are points in the state space where the dynamics of the system are not well-defined, vortex points are singular points where the state trajectories form closed orbits around the point without spiraling inwards or outwards, and saddle points are singular points where the state trajectories approach or diverge from the point in different directions and the system's behavior is unstable and sensitive to small perturbations.
Q10	State the condition/s for a scalar function $V(x)$ to be (i) positive definite (ii) negative definite and (iii) indefinite. Give one example for each
Ans	A scalar function $V(x)$ is said to be: (i) Positive definite if $V(x) > 0$ for all $x \neq 0$, and $V(0) = 0$. In other words, the function is strictly positive for all non-zero values of x and zero at the origin. This implies that the system is stable in the sense that it converges to the origin, and any small perturbations from the equilibrium point decay over time. Example: $V(x) = x^2$ is positive definite. (ii) Negative definite if $V(x) < 0$ for all $x \neq 0$, and $V(0) = 0$. In other words, the function is strictly negative for all non-zero values of x and zero at the origin. This implies that the system is stable in the sense that it converges to the origin, but any small perturbations from the equilibrium point grow over time. Example: $V(x) = -x^2$ is negative definite. (iii) Indefinite if $V(x)$ takes both positive and negative values for different non-zero values of x, and $V(0) = 0$. In other words, the function can be both positive and negative for different values of x, and zero at the origin. This implies that the system is unstable and the behavior depends on the initial conditions. Example: $V(x) = x^3$ is indefinite.

	In summary, a scalar function $V(x)$ is positive definite if it is strictly positive for all non-zero values of x and zero at the origin, negative definite if it is strictly negative for all non-zero values of x and zero at the origin, and indefinite if it takes both positive and negative values for different non-zero values of x and zero at the origin.
PART B	
<i>Answer any one full question from each module, each carries 14 marks.</i> Module I	
Q11	a) From basics, derive the state model of an armature controlled dc motor. b) Differential equation of the system is given by, $\frac{d^3 y}{dt^3} + 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 6y + u = 0$ where, y is the output and u is the input. Derive the state model of the system using phase variables
Ans	a) State model of an armature controlled dc motor: KVL equation for armature circuit: $V_a = I_a R_a + L_a \frac{di_a}{dt} + E_b \quad \left(E_b \propto \frac{d\theta}{dt} \right)$ Torque equation: $T \propto I_a \quad (T = K_T I_a)$ Eqn. for electro mechanical system: $J \frac{d^2 \theta}{dt^2} + B \frac{d\theta}{dt} = T$ Select suitable state variables and output variables, substitute them in the equations, rearrange the terms. Final state equation Final output equation

	b) State model using phase variables  $x_1 = y$ $x_2 = \frac{dy}{dt} = \dot{x}_1$ $x_3 = \frac{d^2y}{dt^2} = \dot{x}_2$ $\therefore \frac{d^3y}{dt^3} + 6\frac{d^2y}{dt^2} + 11\frac{dy}{dt} + 6y + u = 0 \Rightarrow \dot{x}_3 = -6x_3 - 11x_2 - 6x_1 + u$ $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} u \quad [y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
Q12	a) What is similarity transformation? How similarity transformation is helpful for state space analysis? Obtain the diagonal canonical form of the state model given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} u \quad [y] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ b)
Ans	a) Similarity transformation is a linear transformation of a matrix in which the new matrix is obtained by multiplying the original matrix by an invertible matrix on both sides. Specifically, if A is a square matrix and P is an invertible matrix, then the similarity transformation of A by P is given by $P^{-1}AP$. Similarity transformations are useful in state space analysis because they help to identify the eigenvalues and eigenvectors of a system. The eigenvalues of a system are important

because they describe the stability and behavior of the system over time, and the eigenvectors correspond to the directions in which the system responds most strongly.

By performing a similarity transformation on the state space matrix A, it is possible to transform the system into a new coordinate system in which the new matrix A' is diagonal. The diagonal entries of A' are the eigenvalues of A, and the columns of the matrix P that is used to perform the transformation correspond to the eigenvectors of A. This diagonalization process is known as diagonalization by similarity transformation.

Once the state space matrix has been diagonalized, it is much easier to analyze the behavior of the system. In particular, the diagonal entries of the transformed matrix A' correspond to the decay rates of the system, and the eigenvectors of A' provide insight into the direction and strength of the system's response to different inputs. Overall, similarity transformations are a powerful tool for analyzing the behavior of complex systems described by state space models

b

Canonical form

$$[\lambda I - A] = \begin{bmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 6 & 11 & \lambda + 6 \end{bmatrix} \quad \lambda_1 = -1 \quad \lambda_2 = -2 \quad \lambda_3 = -3$$

$$M = [m_1 \quad m_2 \quad m_3] = \begin{bmatrix} 6 & 3 & 2 \\ -6 & -6 & -6 \\ 6 & 12 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \\ 1 & 4 & 9 \end{bmatrix}$$

$$A_{new} = M^{-1} \cdot A \cdot M = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad B_{new} = M^{-1} \cdot B = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$C_{new} = C \cdot M = [1 \quad 1 \quad 1]$$

Module 2

Q13	A discrete time system is given by, $y(k+2) + 5y(k+1) + 6y(k) = u(k)$ where, y is the output and u is the input. Assume zero initial conditions and the sampling period as 1 second. (i) Derive the state model of the system in Jordan canonical form (ii) Draw the block diagram of the state model and (iii) Compute the state transition matrix.
Ans	$z^2 Y(z) + 5zY(z) + 6Y(z) = U(z) \quad \therefore \frac{Y(z)}{U(z)} =$ On solving, $\frac{Y(z)}{U(z)} = \frac{-1}{(z+3)} + \frac{1}{(z+2)}$ $\begin{aligned} x_1(k+1) &= -3x_1(k) + u(k) \\ x_2(k+1) &= -2x_2(k) + u(k) \\ y(k) &= -x_1(k) + x_2(k) \end{aligned}$ $\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} u(k) \\ u(k) \end{bmatrix}$ $y(k) = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$ Block diagram $\begin{aligned} A^k &= Z^{-1} \left\{ [zI - A]^{-1} . z \right\} \\ &= Z^{-1} \left\{ \begin{bmatrix} \frac{1}{(z+3)} & 0 \\ 0 & \frac{1}{(z+2)} \end{bmatrix} z \right\} \\ &= \begin{bmatrix} -3^k & 0 \\ 0 & -2^k \end{bmatrix} \end{aligned}$
Q14	a) State equation of a system is given below. Compute the state transition matrix using Laplace transform method. Also, compute the solution of the given state equation if the initial state vector is $X(0)$. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad X(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

	b) Find the state transition matrix using Cayley-Hamilton theorem for the system matrix given below. $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$
Ans	a) STM using Laplace transform : $[sI - A] = \begin{bmatrix} (s-1) & 0 \\ -1 & (s-1) \end{bmatrix}$ $STM = e^{At} = L^{-1} \{ [sI - A]^{-1} \}$ $= \begin{bmatrix} e^t & 0 \\ te^t & e^t \end{bmatrix}$ $X(t) = e^{At} \cdot X(0)$ $= \begin{bmatrix} e^t \\ te^t \end{bmatrix}$ b) STM using Cayley-Hamilton Theorem $ \lambda I - A = \begin{vmatrix} \lambda & -1 \\ 2 & \lambda + 3 \end{vmatrix} \quad \lambda_1 = -1 \text{ \& } \lambda_2 = -2$ $f(\lambda_1) = \alpha_0 + \alpha_1 \lambda_1 = e^{\lambda_1 t} = e^{-t}$ $f(\lambda_2) = \alpha_0 + \alpha_1 \lambda_2 = e^{\lambda_2 t} = e^{-2t}$ $\therefore \alpha_0 = 2e^{-t} - e^{-2t} \quad \& \quad \alpha_1 = e^{-t} - e^{-2t}$ $f(A) = e^{At} = \alpha_0 \cdot I + \alpha_1 \cdot A = \begin{bmatrix} (2e^{-t} - e^{-2t}) & (e^{-t} - e^{-2t}) \\ (-2e^{-t} + 2e^{-2t}) & (-e^{-t} + 2e^{-2t}) \end{bmatrix}$
Module 3	
Q15	

	a) Using Kalman's method, check the observability of the system given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -3 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad [y] = [40 \quad 10 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ b) Design the observer feed back gain matrix for the desired eigen values at -4, -3±j1 for the state model of the system given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} u \quad [y] = [0 \quad 0 \quad 1] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$
Ans	a) Kalman's method for observability Composite matrix : $Q_0 = \begin{bmatrix} C^T & A^T.C^T & (A^T)^2.C^T \end{bmatrix}$ $Q_0 = \begin{bmatrix} 40 & 0 & 0 \\ 10 & 40 & -30 \\ 0 & 10 & 0 \end{bmatrix}$ b)

Observability matrix, $\mathcal{O}$:

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CAA \end{bmatrix}$$

$$\mathcal{O} = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 0 \\ 3 & 8 & 0 \\ 1 & 1 & 2 \\ 3 & 4 & 8 \\ 4 & 5 & 10 \\ 1 & 2 & 0 \\ 2 & 4 & 0 \end{bmatrix}$$

Rank of $\mathcal{O}$ is 3. Therefore, the system is observable.

$$s^3 + 10s^2 + 29s + 24 = (s + 4)(s + 3 + j)(s + 3 - j)$$

The observer pole locations are -4, -3+j, and -3-j.

The observer feedback gain matrix can be obtained using the formula:

$$L = \begin{bmatrix} L_1 \\ L_2 \\ L_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}^{-1} \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix}$$

where

$$b_0 = 24$$

$$b_1 = 29 - 10L_1$$

$$b_2 = 10L_1 - L_1^2 - 10L_2 - 10L_3$$

Substituting the desired eigenvalues, we get:

	$b_0 = 24$ $b_1 = 101$ $b_2 = 262$ Solving for L, we get: $L = \begin{bmatrix} 4.9874 \\ -9.9487 \\ -3.0000 \end{bmatrix}$ Therefore, the observer feedback gain matrix $K = [k_1 \quad k_2 \quad k_3] = [L_1 \quad L_2 \quad L_3] =$ Observer gain matrix $G = \begin{bmatrix} 25.2 \\ 21.5 \\ 10 \end{bmatrix}$
Q16	a) Using PBH test, check the controllability of the system given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad [y] = [2 \quad 2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ b) Design a feed back controller with state feed back so that the closed loop poles are at -2, -1±j1. The state equation of the original systems is given below. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} u$
Ans	a

PBH method for controllability : condition $\text{rank} \left\{ \begin{bmatrix} (\lambda I - A) & B \end{bmatrix} \right\} = n$

$$n = 2 \quad \lambda_1 = -1 \quad \lambda_2 = -4$$

For λ_1 : $\text{rank} \left\{ \begin{bmatrix} -1 & -1 & 0 \\ 4 & 4 & 1 \end{bmatrix} \right\} = 2$

For λ_2 : $\text{rank} \left\{ \begin{bmatrix} -4 & -1 & 0 \\ 4 & 1 & 1 \end{bmatrix} \right\} = 2$

System is completely controllable.

b

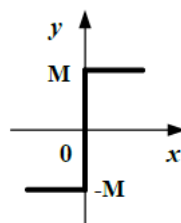
Feed back controller with a state feed back:

State feedback gain matrix: $K = [0.4 \quad 0.4 \quad 0.1]$

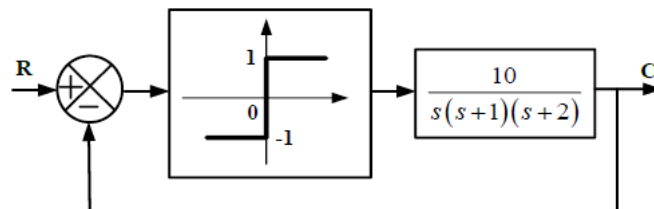
Module 4

Q17

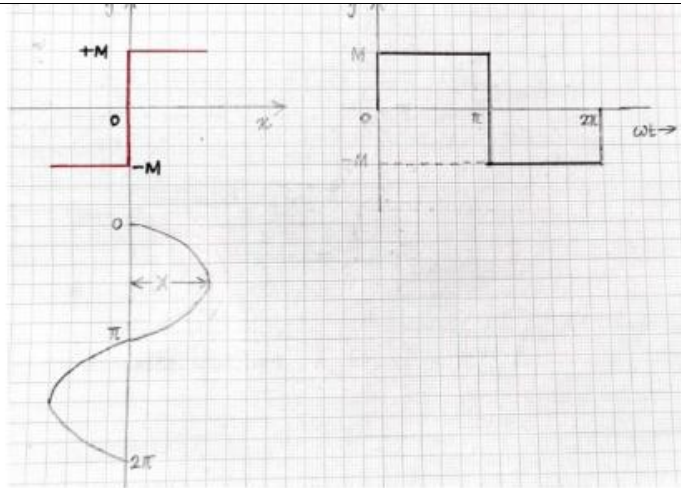
a) Derive the describing function of ideal relay non-linearity given below.



b) Determine the frequency and nature of the limit cycle for the unity feed back system given below.



Ans



Input signal: $x = X \sin \omega t$

Output signal:

$$y(\omega t) = \begin{cases} +M & ; 0 < \omega t < \pi \\ -M & ; \pi < \omega t < 2\pi \end{cases}$$

Describing function :

$$A_1 = \frac{4M}{\pi} \quad B_1 = 0 \quad \phi_1 = 0^\circ$$

Derivation of constants with necessary steps

$$K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{(A_1^2 + B_1^2)} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$$

$$K_N(X, \omega) = \frac{4M}{\pi X} \angle 0^\circ$$

b

Frequency and nature of the limit cycle:

Plot of $G(j\omega)$

$$\frac{-1}{K_N(X, \omega)} \text{ locus}$$

$$G(j\omega) = \frac{10}{s(s+1)(s+2)} = \frac{5}{j\omega(1+j\omega)(1+j0.5\omega)}$$

$$|G(j\omega)| = \frac{5}{\omega \sqrt{(1+\omega^2)} \sqrt{(1+0.25\omega^2)}} \quad \angle G(j\omega):$$

$$\frac{-1}{K_N(X, \omega)} = \frac{\pi X}{4} \angle -180^\circ$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega)$$

ω rad/s		
$ G(j\omega) $		
$\angle G(j\omega)$		-180°

$G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180° axis

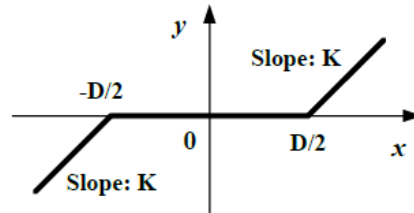
$$-90 - \tan^{-1}(\omega) - \tan^{-1}(0.5\omega) = -180^\circ$$

$$\therefore \omega = \sqrt{2} \text{ rad / s}$$

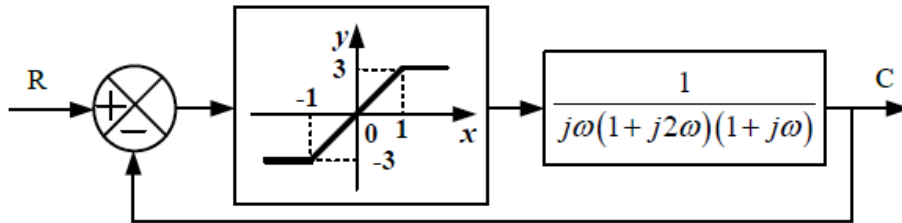
Nature of the limit cycle: stable

Q18

a) Derive the describing function of dead-zone non-linearity given below.



b) Determine the frequency and nature of the limit cycle for the unity feed back system given below.

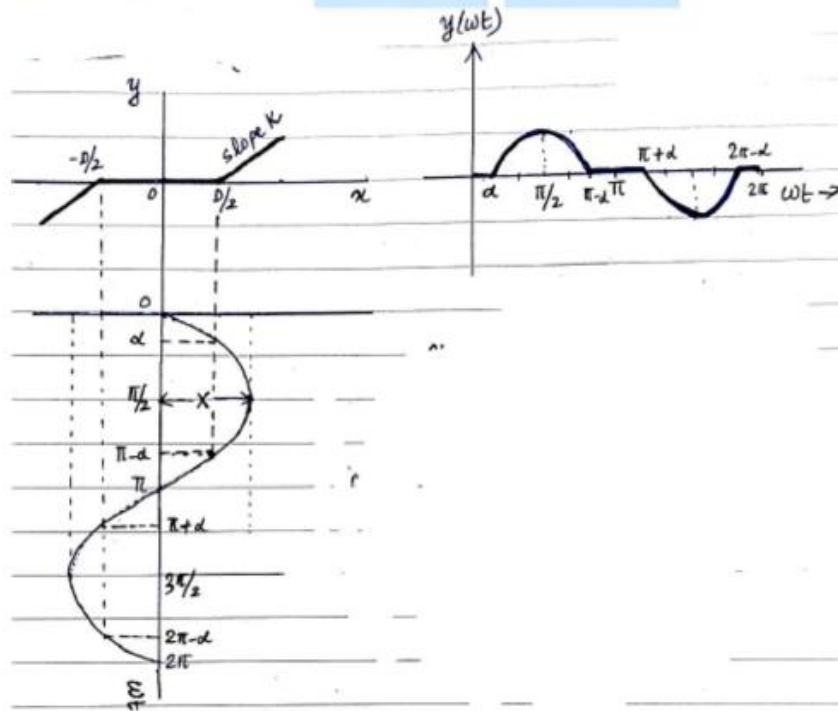


Ans

a

Describing function of dead-zone non-linearity:

Waveforms :



Input signal: $x = X \sin \omega t$

Output signal:

$$y(\omega t) = \begin{cases} 0 & ; 0 < \omega t < \alpha \\ K \left(x - \frac{D}{2} \right) & ; \alpha < \omega t < (\pi - \alpha) \\ 0 & ; (\pi - \alpha) < \omega t < \pi \end{cases}$$

Describing function :

$$K_N(X, \omega) = \frac{Y_1}{X} \angle \phi_1 \quad Y_1 = \sqrt{A_1^2 + B_1^2} \quad \phi_1 = \tan^{-1} \left(\frac{B_1}{A_1} \right)$$

$$A_1 = \frac{4K}{\pi} \left[\frac{X}{2} \left(\frac{\pi}{2} - \alpha + \frac{\sin 2\alpha}{2} \right) - \frac{D}{2} \cos \alpha \right] = KX \left[1 - \frac{2}{\pi} (\alpha + \sin \alpha \cos \alpha) \right]$$

$$B_1 = 0 \quad \phi_1 = 0^\circ$$

b

Frequency and nature of the limit cycle:

Plot of $G(j\omega)$

$$\frac{-1}{K_N(X, \omega)} \text{ locus}$$

$$G(j\omega) = \frac{1}{j\omega(1+j2\omega)(1+j\omega)}$$

$$|G(j\omega)| = \frac{1}{\omega\sqrt{(1+4\omega^2)}\sqrt{(1+\omega^2)}} \quad \angle G(j\omega) = -90^\circ$$

$$\frac{-1}{K_N(X, \omega)} = \begin{cases} \frac{\pi}{2K\pi[\beta + \sin\beta \cdot \cos\beta]} \angle -180^\circ & ; X > 1 \\ \frac{1}{K} \angle -180^\circ & ; X < 1 \end{cases}$$

$$\angle G(j\omega) = -90^\circ - \tan^{-1}(2\omega) - \tan^{-1}(\omega)$$

ω rad/s		0.707
$ G(j\omega) $		
$\angle G(j\omega)$		-180°

$G(j\omega)$ locus and $\frac{-1}{K_N(X, \omega)}$ locus intersect at -180° axis

$$-90 - \tan^{-1}(2\omega) - \tan^{-1}(\omega) = -180^\circ$$

$$\therefore \omega = 0.707 \text{ rad/s}$$

Nature of the limit cycle: stable

Module 5

Q19	a) What do you mean by phase trajectory? Explain how to draw the phase trajectory using isocline method. b) Compute the Lyapunov function, $V(X)$ for which the system given below is asymptotically stable. $\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$
Ans	a) A phase trajectory, also known as a phase plot or phase portrait, is a graphical representation of the behavior of a dynamical system in state space. It shows how the state variables of the system evolve over time, and how they depend on each other. To draw a phase trajectory using the isocline method, follow these steps: Step 1: Determine the system's equilibrium points. An equilibrium point is a state where the system remains unchanged over time. It is the point where the derivative of the state variables is zero. To find the equilibrium points, set the derivative equations of the state variables to zero and solve for the state variables. Step 2: Draw the isoclines. An isocline is a curve in state space along which the derivative of one of the state variables is constant. To draw the isoclines, set the derivative of one state variable to a constant value and plot the resulting curve in state space. Step 3: Determine the direction of the arrows. The direction of the arrows on the phase trajectory indicates the direction in which the state variables change over time. The direction of the arrows can be determined by examining the sign of the derivative of the state variables along the isoclines. Step 4: Sketch the phase trajectory. Sketch the phase trajectory by connecting the equilibrium points and the isoclines. The phase trajectory shows the qualitative behavior of the system in state space, including the stability and the types of oscillations or transient behavior. The isocline method is a useful technique for drawing phase trajectories of nonlinear systems with two state variables. However, it may not be applicable for more complex systems or systems with more than two state variables. In such cases, numerical methods such as simulation or bifurcation analysis may be used. b)

	Lyapunov function: $-Q = A^T \cdot P + P \cdot A$ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} -1 & -2 \\ 1 & -4 \end{bmatrix}$ On solving, $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 23/60 & -7/60 \\ -7/60 & 11/60 \end{bmatrix}$ Positive definiteness of P for stability. $ p_{11} > 0$ $\begin{vmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{vmatrix} = \frac{17}{300} > 0$ Lyapunov function: $V(X) = X^T \cdot P \cdot X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}^T \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{23}{60} x_1^2 - \frac{14}{60} x_1 x_2 + \frac{11}{60} x_2^2$
Q20	a) Construct the phase trajectory of the second order system given below using isocline method. $\dot{x}_2 + (2\xi\omega_n)x_2 + (\omega_n^2)x_1 = 0$ $X(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \end{bmatrix} \quad \xi = 0.15 \quad \omega_n = 1 \text{ rad/s}$ where x_1 and x_2 are the state variables. b) Explain Lyapunov function? Using Lyapunov function, $V(X)$, state the condition/s for the system to be (i) stable (ii) asymptotically stable in the large and (iii) unstable. Consider the system described by, $\dot{x}_1 = x_2$ $\dot{x}_2 = -x_1 - x_2^3$ Check the stability of the system for the Lyapunov function, $V(X) = x_1^2 + x_2^2$.

Ans	a To construct the phase trajectory of the second-order system using the isocline method, we first need to find the equilibrium points and draw the isoclines. Equilibrium points: To find the equilibrium points, we set the derivative equations of the state variables to zero and solve for the state variables. For the given system, the state variables are x_1 and x_2. Taking their derivatives with respect to time gives: $\frac{dx_1}{dt} = \xi_1 x_1 + \xi_2 x_2 + \xi_3 = 0$ Setting these equations to zero gives: $x_1 = 0, x_2 = 0$ Therefore, the equilibrium point is (0,0). Isoclines: An isocline is a curve in state space along which the derivative of one of the state variables is constant. For the given system, we can find the isoclines by setting the derivative of each state variable to a constant value and solving for the other variable. Isocline for x_1: $\frac{dx_1}{dt} = \xi_1 x_1 + \xi_2 x_2 + \xi_3 = 0$ Setting the derivative of x_1 to a constant value, say k, we get: $\xi_1 x_1 + \xi_2 x_2 + \xi_3 = k$ This is a quadratic equation in x_2, and we can solve for x_2 using the quadratic formula: $x_2 = \frac{-\xi_2 \pm \sqrt{\xi_2^2 - 4\xi_1(\xi_3 - k)}}{2\xi_1}$ So the isocline for x_1 is a pair of straight lines given by: $x_2 = \frac{-\xi_2 \pm \sqrt{\xi_2^2 - 4\xi_1(\xi_3 - k)}}{2\xi_1}$ Isocline for x_2: $\frac{dx_2}{dt} = \xi_4 x_1 + \xi_5 x_2 + \xi_6 = 0$ Setting the derivative of x_2 to a constant value, say k, we get: $\xi_4 x_1 + \xi_5 x_2 + \xi_6 = k$ This is also a quadratic equation in x_1, and we can solve for x_1 using the quadratic formula: $x_1 = \frac{-\xi_4 \pm \sqrt{\xi_4^2 - 4\xi_5(\xi_6 - k)}}{2\xi_5}$ So the isocline for x_2 is a pair of straight lines given by: $x_1 = \frac{-\xi_4 \pm \sqrt{\xi_4^2 - 4\xi_5(\xi_6 - k)}}{2\xi_5}$ Direction of arrows: The direction of arrows on the phase trajectory indicates the direction in which the state variables change over time. We can determine the direction of arrows by examining the sign of the derivative of the state variables along the isoclines. For the given system, the derivative of x_1 is negative along the isoclines for x_1, and the derivative of x_2 is negative along the isocline
	b Let X be the state vector of a dynamic system and $V(X)$ be a scalar function defined over the state space of the system. $V(X)$ is a Lyapunov function if the following conditions are satisfied: $V(X)$ is positive definite, i.e., $V(X) > 0$ for all $X \neq 0$. $V(X)$ is continuous in the state space. The time derivative of $V(X)$ along the trajectories of the system is negative semi-definite, i.e., $dV(X)/dt \leq 0$ for all $X \neq 0$. Using a Lyapunov function, we can determine whether a system is stable, asymptotically stable, or unstable. (i) Stable: A system is said to be stable if for any initial state X_0, the trajectories of the system converge to a bounded region in the state space. A Lyapunov function $V(X)$ can be used to show that a system is stable if it is positive definite and $dV(X)/dt \leq 0$ for all $X \neq 0$.

	(ii) Asymptotically stable in the large: A system is said to be asymptotically stable if for any initial state X_0, the trajectories of the system converge to a unique equilibrium point in the state space, and this equilibrium point is globally attractive. A Lyapunov function $V(X)$ can be used to show that a system is asymptotically stable in the large if it is positive definite, $dV(X)/dt \leq 0$ for all $X \neq 0$, and $V(X) \rightarrow 0$ as $\ X\ \rightarrow \infty$. (iii) Unstable: A system is said to be unstable if for some initial states X_0, the trajectories of the system diverge to infinity in the state space. A Lyapunov function $V(X)$ can be used to show that a system is unstable if there exists at least one X_0 such that $V(X_0) > 0$ and $dV(X)/dt > 0$ for all $X \neq 0$. Now, let's consider the given system: We want to check the stability of the system for the Lyapunov function $V(X) = x_1^2$. Taking the time derivative of $V(X)$, we have: $dV(X)/dt = 2x_1 dx_1/dt$ Substituting the dynamics of the system, we get: $dV(X)/dt = -2x_1^2 - 4x_1 x_2$ Since $dV(X)/dt$ is negative definite, the system is stable. Note that we cannot determine whether the system is asymptotically stable or not using this Lyapunov function
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